

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1 : Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Application-Based Questions

Weightage: 40%

QUESTIONS:

Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.

The given code contains only a `<form>` element without any semantic HTML5 tags. To improve the page structure, elements such as `<header>` for the page title, `<main>` for the primary content, and `<article>` or `<section>` for the registration form should be included. The form should also use `<label>` elements for each input field so that users can easily understand what information needs to be entered. These semantic elements improve accessibility, maintainability, and search engine indexing.

2. Modify the structure to make it **standards-compliant**.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Symposium Portal</title>
</head>
<body>

  <header>
    <h1>College Symposium Registration</h1>
  </header>

  <main>
    <article>
      <form action="/register" method="POST">
        <label for="studentName">Student Name</label>
        <input type="text" id="studentName" name="name" required>

        <label for="studentEmail">Email Address</label>
        <input type="email" id="studentEmail" name="email" required>

        <button type="submit">Register</button>
      </form>
    </article>
  </main>

</body>
</html>
```

3. Interpret the HTTP request generated during form submission.

When the registration form is submitted, the browser creates an HTTP **POST** request and sends it to the `/register` URL. The request contains the student's name and email address in the request body along with HTTP headers such as the content type. The server receives this request and processes the registration details without exposing the submitted data in the browser's address bar.

4. Analyze the HTTP response cycle from server to browser.

After the server receives the registration request, it verifies the submitted details and performs the required processing, such as saving the student's information in a database. If the registration is successful, the server sends a success response, usually with the **200 OK** status code, along with a confirmation page. If an error occurs during processing, the server returns an appropriate error message, which the browser displays to the user.

5. Suggest improvements for usability and accessibility.

The form can be enhanced by adding descriptive labels, input placeholders, and validation messages to guide users while filling in the form. Using semantic HTML elements improves compatibility with screen readers. Keyboard navigation should be supported so users can access all controls without a mouse. The form should also be responsive for different screen sizes and use sufficient colour contrast to improve readability. These improvements provide a better user experience and make the webpage more accessible to all users.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.

The HTML code contains structural errors because the `<h1>` and `<p>` elements are not properly closed. The `<p>` tag is placed before closing the `<h1>` tag, resulting in incorrect nesting. Although modern browsers try to fix such errors automatically, the HTML is not valid according to W3C standards. Properly closing all tags ensures that the webpage is displayed consistently across different browsers.

2. Analyze how different browsers may interpret this structure.

Different browsers have different rendering engines, so they may handle invalid HTML in different ways. One browser may automatically close the missing tags and display the page correctly, while another browser may show incorrect spacing or formatting. This can lead to inconsistent layouts and affect the user experience. Writing valid HTML helps avoid these browser-specific differences.

3. Identify **missing semantic elements** (e.g., header, section).

The webpage does not use HTML5 semantic elements that improve the document structure. A `<header>` element should contain the welcome heading, while a `<main>` element should hold the main content. The "About us" information should be placed inside a `<section>` element. These semantic tags improve accessibility, code organization, and search engine optimization (SEO).

4. Provide a **corrected W3C-compliant HTML structure**.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Company</title>
</head>
<body>

<header>
  <h1>Welcome</h1>
</header>

<main>
  <section>
    <p>About us</p>
  </section>
</main>

</body>
</html>
```

5. Suggest techniques to ensure **cross-browser compatibility**.

Cross-browser compatibility can be achieved by writing valid HTML and CSS according to W3C standards and testing the website in browsers such as Chrome, Firefox, Edge, and Safari. Developers should use CSS Reset or Normalize.css to reduce browser styling differences. Responsive design, standard web technologies, and avoiding deprecated HTML elements also help maintain a consistent layout. Regular validation and browser testing ensure that the website works correctly for all users.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.

The form contains two important mistakes. The button is defined as `type="button"` instead of `type="submit"`, so clicking it does not submit the form. Also, the form uses the **GET** method to send login details, which is not suitable for passwords because the data is added to the URL. These issues prevent the form from working correctly and reduce security.

2. Analyze why the **HTTP request is not triggered**.

The HTTP request is not triggered because the button is not a submit button. A button with `type="button"` performs no action unless it is connected to JavaScript. Since no JavaScript is provided to submit the form, clicking the button does nothing. As a result, the browser does not send any request to the server.

3. Interpret the role of **GET method in this scenario**.

The **GET** method is used to retrieve information from the server by sending form data as part of the URL. It is suitable for search forms and non-sensitive data. In this login form, using GET is not appropriate because the username and password may be visible in the browser's address bar, browser history, and server logs. Therefore, the **POST** method should be used instead.

4. Propose a **corrected implementation**.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Login Form</title>
</head>
<body>

  <form action="/submit" method="POST">
    <label for="username">Username</label>
    <input type="text" id="username" name="username" required>

    <label for="password">Password</label>
    <input type="password" id="password" name="password" required>

    <button type="submit">Login</button>
  </form>

</body>
</html>
```

5. Suggest improvements for **secure data transmission**.

The application should use the **POST** method to prevent sensitive information from appearing in the URL. The website should also use **HTTPS** so that data is encrypted during transmission. Input validation should be performed on both the client and server sides, and passwords should be stored in encrypted or hashed form. Additional security measures such as CSRF protection and strong password policies can further improve the security of the login process.

Code of Conduct:

I (K .sai kumar / 192372080) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

k.sai kumar

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand